WATER LEAKAGE DETECTION USING MACHINE LEARNING

Abstract

This project presents a machine learning-based approach for detecting water leakage in a network of 100 connected water tanks.

The dataset contains 100,000 timestamps and 301 columns. It provides pressure measurements, refill indicators and binary leakage labels for each tank.

The project implements four classification algorithms: Logistic Regression, Decision Tree, Random Forest and Support Vector Machine (SVM).

A chronological training, validation and testing strategy was adopted. Feature engineering was performed using pressure, pressure change and refill indicators.

The final testing results showed that Random Forest achieved the highest accuracy of 86.55%, while SVM achieved the highest recall of 76.43% and F1-score of 73.32%.

SVM was selected as the preferred model when detecting a greater proportion of actual leakage observations is the primary objective.

A proposed deployment architecture and monitoring strategy were also documented. A simulated inference function was implemented to demonstrate predictions from sensor readings.

1. Introduction

Water leakage is an important problem in water storage and distribution systems. Leakage can result in water wastage, reduced system efficiency and pressure changes.

Manual inspection can be time-consuming.

Machine learning provides an approach for analysing sensor readings and identifying patterns associated with leakage conditions.

This project investigates the use of supervised classification algorithms for detecting leakage in a simulated network of connected water tanks.

2. Problem Statement

Develop a machine learning system that analyses water tank pressure and refill measurements to classify whether a tank is experiencing leakage.

The system should compare multiple classification algorithms and evaluate their ability to identify leakage observations.

3. Objectives

1. Obtain and understand the water tank dataset.
2. Perform exploratory data analysis.
3. Examine the temporal behaviour of leakage.
4. Prepare features for machine learning.
5. Implement four classification algorithms.
6. Evaluate model performance.
7. Investigate feature importance and prediction errors.
8. Document a proposed deployment architecture.
9. Establish monitoring and reproducibility procedures.

4. Literature Review

4.1 Khorasgani et al. (2019)

The researchers investigated fault detection and isolation in industrial networks using Graph Convolutional Neural Networks.

Their work includes a connected water tank network, which is the source of the dataset used in this project.

Their research considers relationships between connected network components. Our implementation instead uses individual tank measurements as tabular classification features.

4.2 Alves Coelho et al. (2020)

The authors presented a machine learning-based approach for water leak detection using real-time sensor data.

Their research highlights the use of wireless sensor networks and automated monitoring for identifying leaks.

4.3 Mashhadi et al. (2021)

The researchers investigated machine learning methods for leak detection and localization in water distribution systems.

Their study examined pressure and flow-related information and compared supervised and unsupervised learning methods.

The research demonstrates the importance of selecting appropriate algorithms for water distribution data.

4.4 Kammoun et al. (2022)

This study presents a comparative survey of artificial intelligence applications for leak detection in water distribution networks.

It provides an overview of different approaches to leak detection and localization.

4.5 Shen and Cheng (2022)

The authors investigated tree-based machine learning methods for pipeline leakage detection.

Their research addresses the challenges of identifying leakage from practical pipeline signals.

4.6 Lee and Kim (2023)

The researchers investigated machine learning approaches for detecting leakage using water pipeline vibration sensor measurements.

Their work demonstrates the application of machine learning to sensor-based leak detection.

5. Dataset Description

Dataset: IndustrialNetwork/GraphDataset

Source: https://github.com/IndustrialNetwork/GraphDataset

Dataset repository commit:

862da8b419e65b240c107a343b72dad44915c811

The repository contains 14 CSV files, including:

- 12 measurement files.
- Tank location information.
- Tank connection matrix.

The 12 measurement files were combined.

Original dataset dimensions:

Parameter	Value
Rows	100,000
Columns	301
Pressure measurements	100
Refill indicators	100
Leak indicators	100
Total tank-time observations	10,000,000

6. Exploratory Data Analysis

6.1 Data Quality

Parameter	Result
Missing values	0
Duplicate rows	0
Minimum timestamp	0.0

Parameter	Result
Maximum timestamp	99999.0
Time ordering	Strictly increasing

6.2 Sensor Statistics

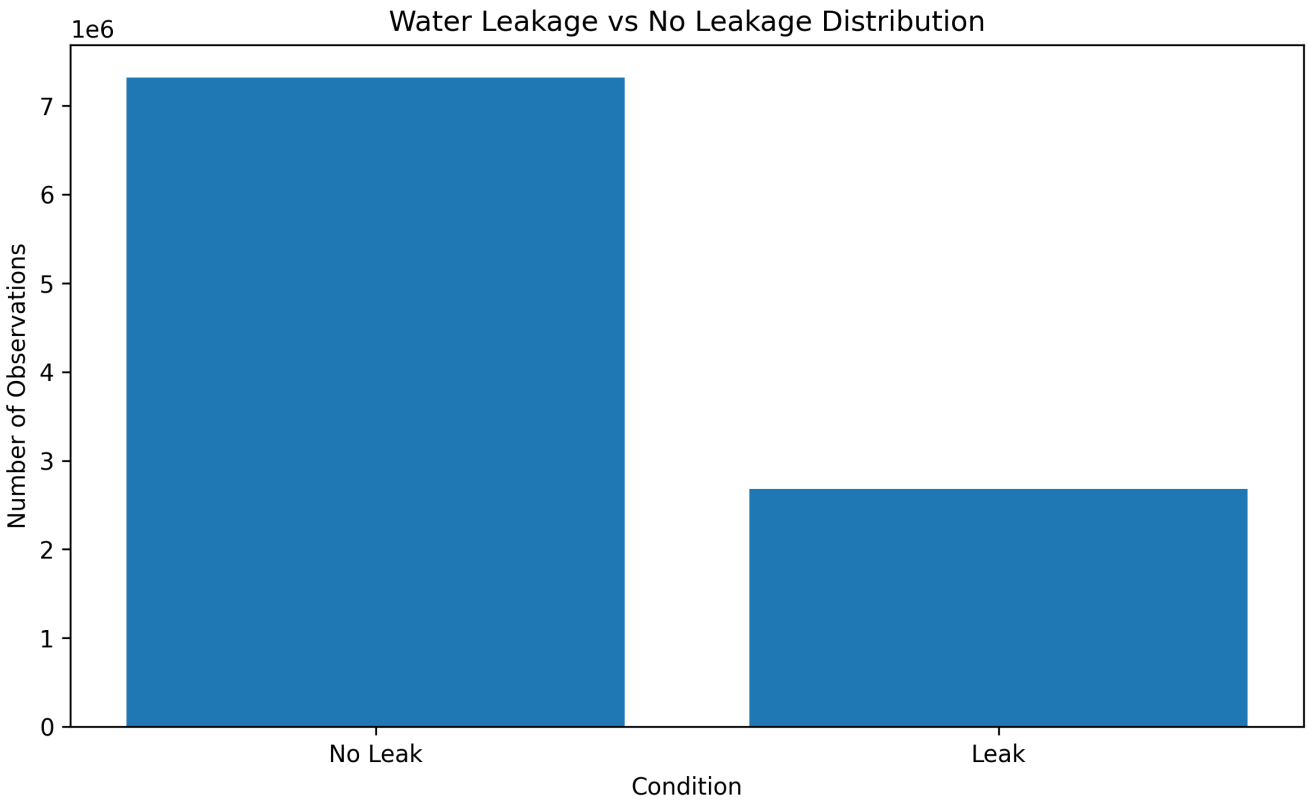
Sensor	Minimum	Maximum	Mean
Pressure	1.2302	102.3437	54.2629
Refill	0	1	0.05998
Leak	0	1	0.26801

The observed maximum pressure value exceeds 100 slightly. This observation was retained and not automatically capped.

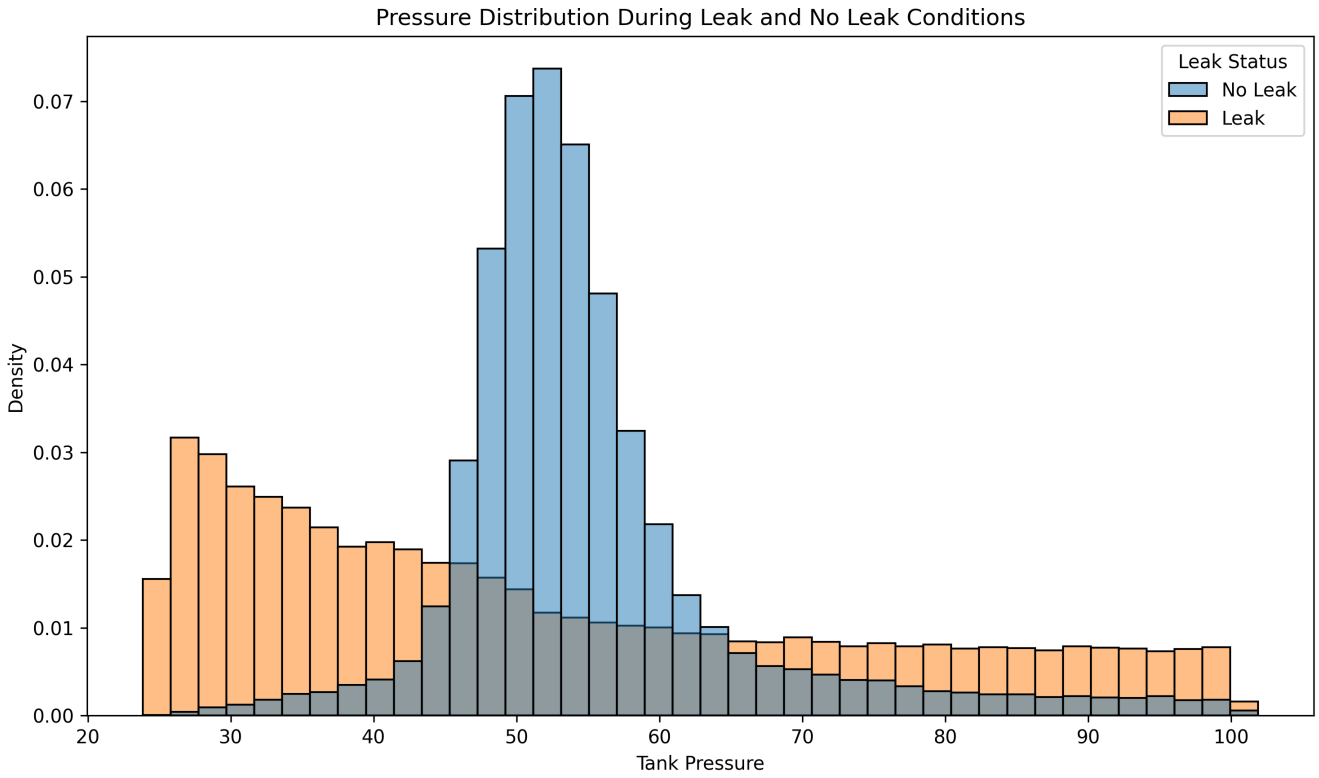
It should be investigated against the dataset's documented sensor range before any production use.

6.3 Leakage Distribution

Condition	Observations	Percentage
No Leak	7,319,875	73.20%
Leak	2,680,125	26.80%



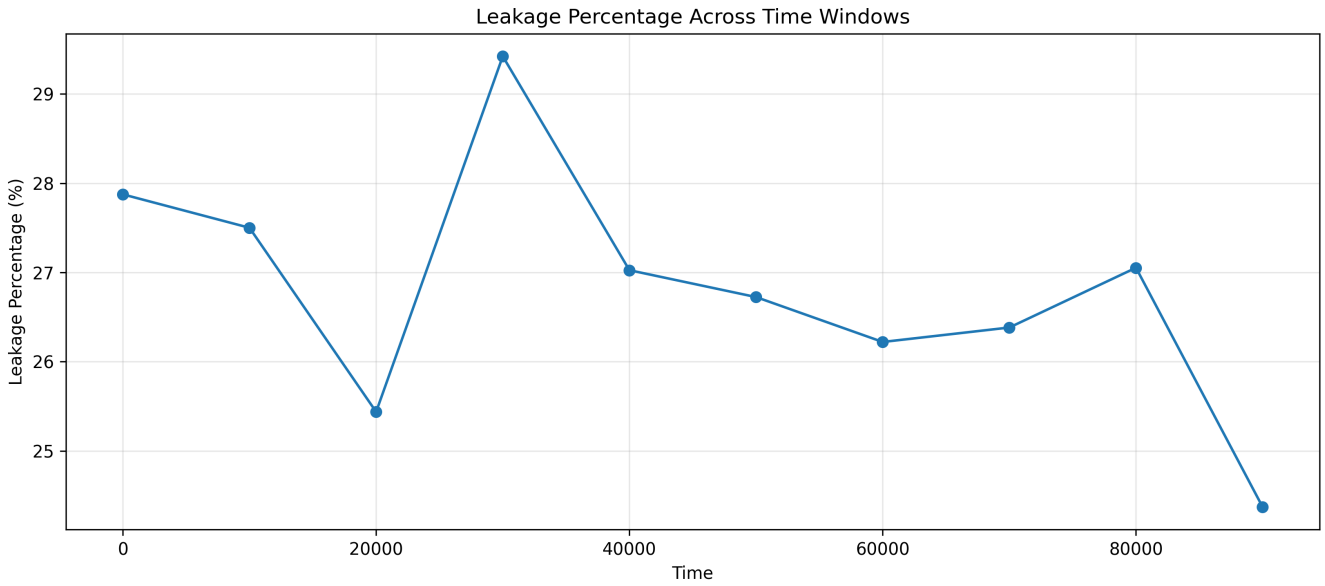
6.4 Pressure Distribution



6.5 Temporal Analysis

Leakage percentages were calculated across ten chronological windows to investigate changes over time.

Start Time	End Time	Leak Percentage	Average Pressure	Refill Percentage
0.0	9999.0	27.875	54.928	6.278
10000.0	19999.0	27.5	53.934	6.127
20000.0	29999.0	25.438	53.763	5.728
30000.0	39999.0	29.421	53.751	6.55
40000.0	49999.0	27.025	54.13	6.06
50000.0	59999.0	26.724	54.645	5.965
60000.0	69999.0	26.221	54.62	5.869
70000.0	79999.0	26.383	53.672	5.894
80000.0	89999.0	27.053	53.949	6.055
90000.0	99999.0	24.372	55.236	5.452



7. Data Preprocessing

The dataset was divided chronologically.

Partition	Timestamp Range	Sample Size
Training	0–69,999	30,000
Validation	70,000–84,999	10,000
Testing	85,000–99,999	10,000

Stratified sampling was applied independently to each chronological partition.

The final modelling dataset contains 50,000 observations.

This approach avoids placing observations from the same timestamp into different partitions.

8. Feature Engineering

Three input features were used:

Feature	Description
Pressure	Current tank pressure
Pressure Change	Difference between current and previous pressure
Refill	Binary refill indicator

Pressure change was calculated as:

Pressure Change = Current Pressure - Previous Pressure

The target variable is Leak:

- 0 = No Leak

- 1 = Leak

Time and Tank_ID were retained for analysis but were not included as model input features.

9. Machine Learning Implementation

Four supervised classification algorithms were implemented.

9.1 Logistic Regression

Used as the baseline linear classification algorithm.

9.2 Decision Tree

Used to identify nonlinear relationships through tree-based decision rules.

9.3 Random Forest

An ensemble model using 200 decision trees.

9.4 Support Vector Machine

An RBF-kernel SVM was implemented to identify nonlinear classification boundaries.

StandardScaler was included in the Logistic Regression and SVM pipelines.

Tree-based models were not scaled.

Class weighting was used to account for the unequal class distribution.

9.5 Model Training Time

Model	Training Time (seconds)
Logistic Regression	0.0583
Decision Tree	0.4034
Random Forest	19.2164
SVM	48.9462

10. Final Model Evaluation

All four trained models were evaluated using the previously held-out test sample of 10,000 observations.

10.1 Evaluation Metrics

Accuracy measures the proportion of correct predictions.

Precision measures how many predicted leaks were actually leakage observations.

Recall measures how many actual leakage observations were detected.

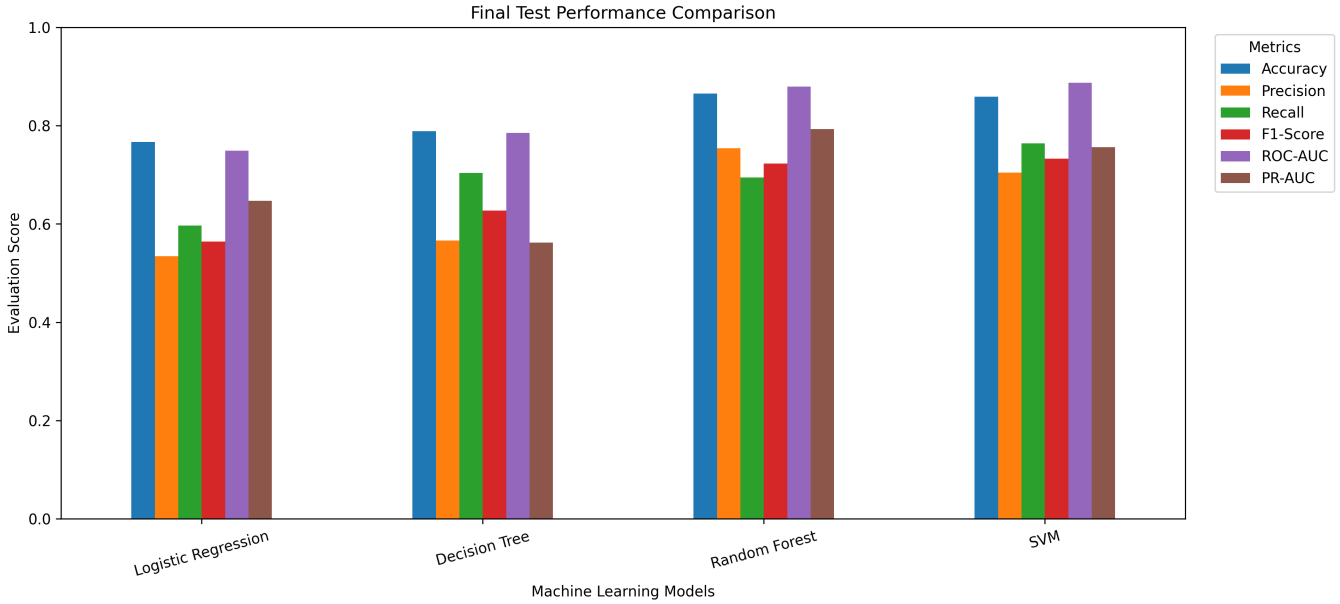
F1-score provides a balance between precision and recall.

ROC-AUC measures the ability to distinguish between the two classes across classification thresholds.

PR-AUC summarizes the precision-recall relationship.

10.2 Final Test Results

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	PR-AUC
Logistic Regression	0.7666	0.5345	0.5967	0.5639	0.749	0.6472
Decision Tree	0.7888	0.5664	0.7034	0.6275	0.7853	0.5622
Random Forest	0.8655	0.7541	0.6947	0.7232	0.8798	0.7928
SVM	0.8593	0.7044	0.7643	0.7332	0.8873	0.7565



11. Confusion Matrix Analysis

The final test confusion matrix counts are:

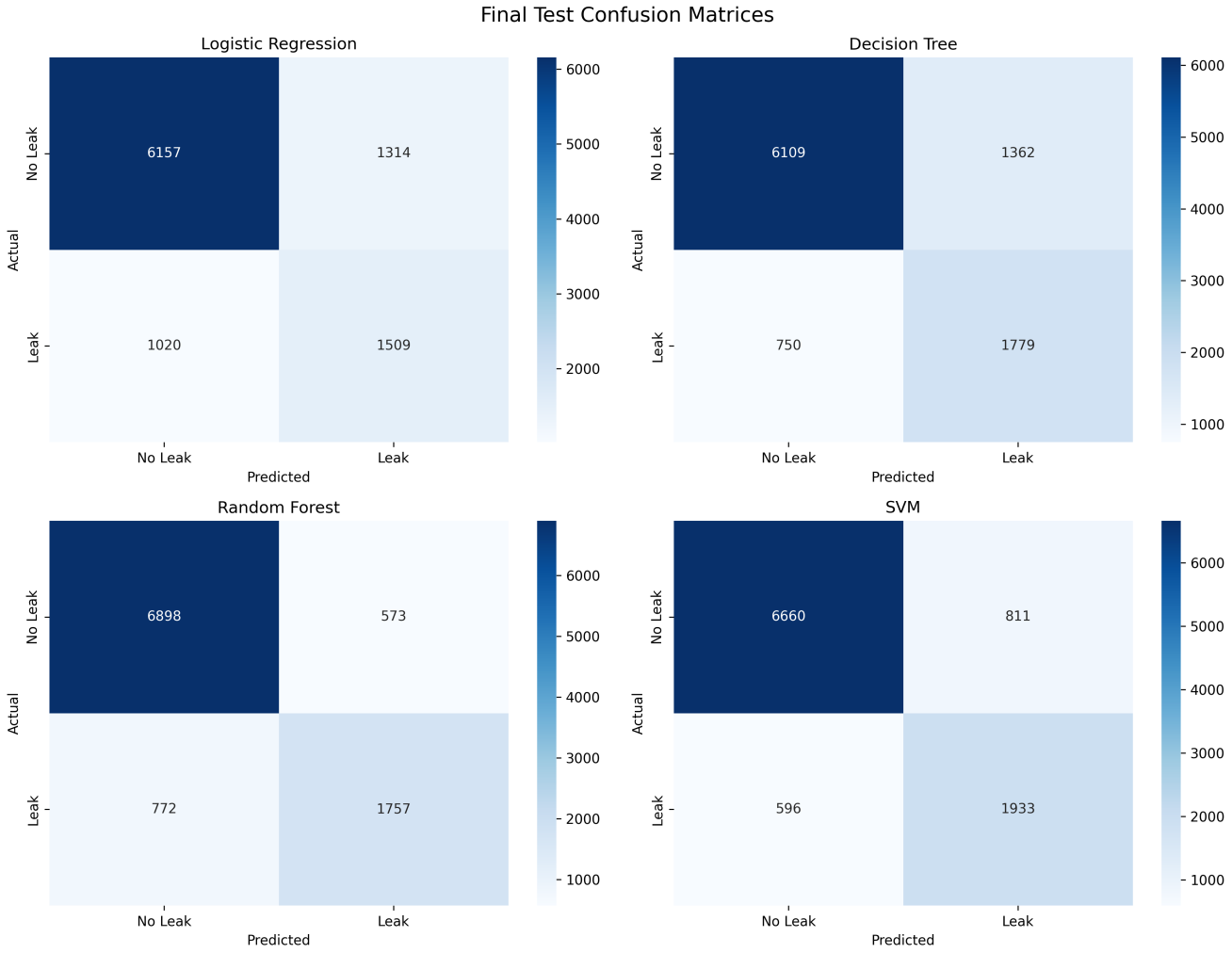
Model	TN	FP	FN	TP
Logistic Regression	6157	1314	1020	1509
Decision Tree	6109	1362	750	1779
Random Forest	6898	573	772	1757
SVM	6660	811	596	1933

TN = True Negative

FP = False Positive

FN = False Negative

TP = True Positive



Random Forest produced fewer false alarms than SVM.

SVM detected more actual leakage observations and produced fewer missed leaks.

12. Comparative Analysis

Random Forest achieved the highest test accuracy of 86.55%.

It also obtained the highest precision of 75.41% and PR-AUC of 79.28%.

SVM achieved the highest recall of 76.43%, F1-score of 73.32% and ROC-AUC of 88.73%.

SVM detected 1,933 actual leaks and missed 596.

Random Forest detected 1,757 actual leaks and missed 772.

SVM is therefore preferred when reducing missed leakage observations is the primary objective.

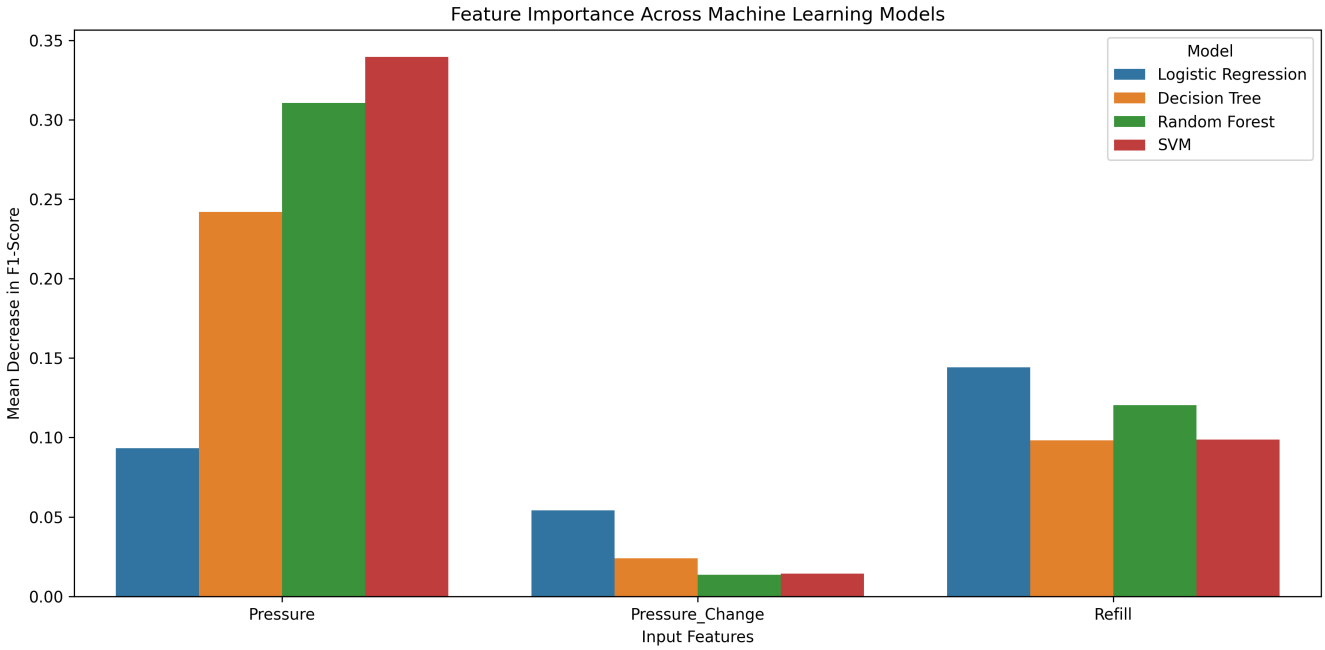
Random Forest remains an alternative when higher precision, fewer false alarms and faster predictions are important.

13. Model Explainability

Permutation importance was used to examine the contribution of individual input features.

The analysis used 2,000 validation observations and five permutation repeats.

Model	Feature	Importance	Standard Deviation
Logistic Regression	Pressure	0.09328	0.01015
Logistic Regression	Pressure_Change	0.0542	0.00651
Logistic Regression	Refill	0.14418	0.00484
Decision Tree	Pressure	0.24193	0.00759
Decision Tree	Pressure_Change	0.02394	0.01257
Decision Tree	Refill	0.0982	0.00453
Random Forest	Pressure	0.31051	0.01461
Random Forest	Pressure_Change	0.0136	0.00704
Random Forest	Refill	0.12026	0.00436
SVM	Pressure	0.33952	0.01954
SVM	Pressure_Change	0.01437	0.00475
SVM	Refill	0.0988	0.00556



Pressure had the greatest predictive importance for SVM, Decision Tree and Random Forest.

Refill was the most important feature for Logistic Regression.

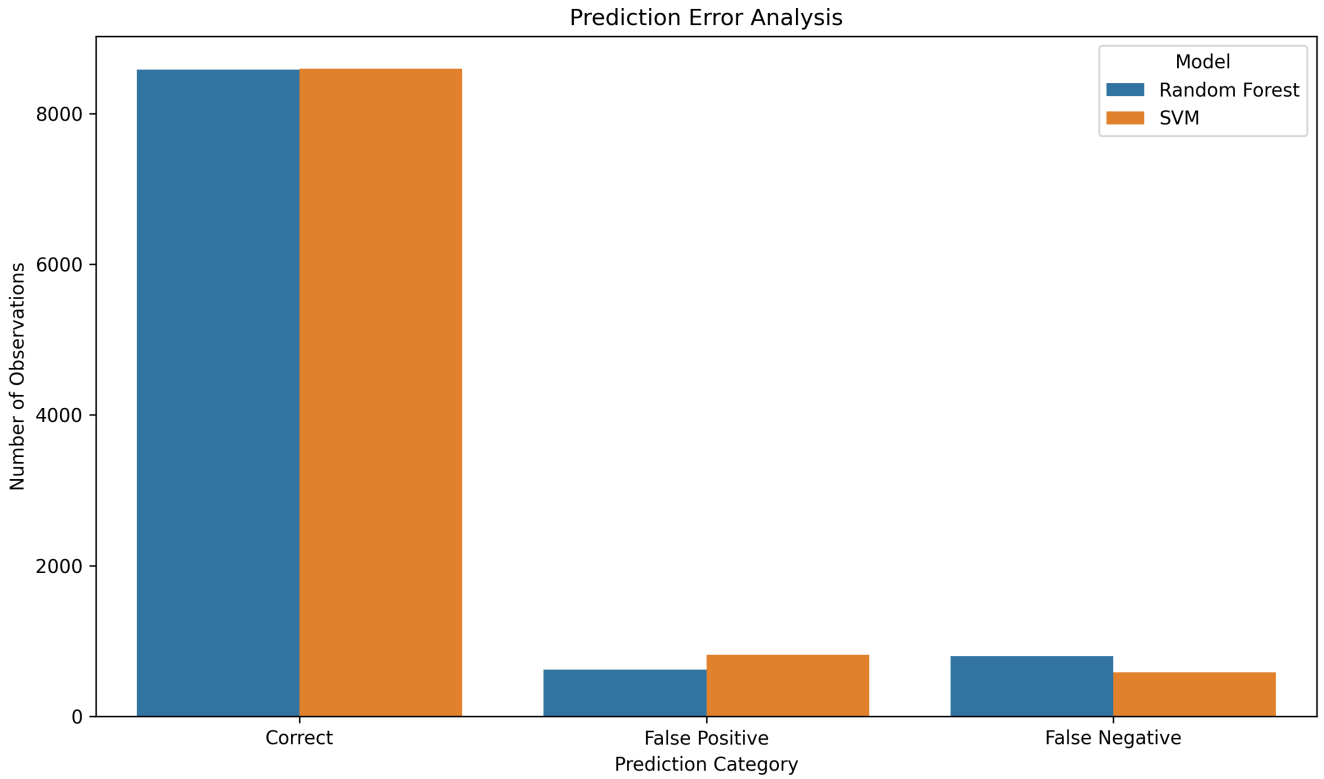
Pressure Change had comparatively low importance across the four models.

Permutation importance measures predictive contribution and does not establish causality.

14. Error Analysis

Error analysis was performed using the validation dataset.

Model	Error Type	Count
Random Forest	Correct	8579
Random Forest	False Positive	621
Random Forest	False Negative	800
SVM	Correct	8593
SVM	False Positive	820
SVM	False Negative	587



The results indicate a trade-off between false alarms and missed leakage observations.

SVM produced fewer missed leaks than Random Forest, while Random Forest generated fewer false alarms.

15. Proposed Deployment Architecture

The proposed architecture follows this workflow:

Water Tank Sensors

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Data Collection

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Data Validation

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Feature Engineering

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Trained SVM Model

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Leak Prediction

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Alert Generation

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Monitoring Dashboard

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Historical Data Logging

A simulated inference function was implemented to demonstrate how pressure and refill observations could be passed to the trained model.

Physical sensor integration and production deployment have not been implemented.

16. Model Monitoring

The proposed monitoring procedures include:

- Accuracy, precision, recall and F1-score.
- False positive and false negative monitoring.
- Sensor data quality checks.
- Pressure and refill distribution monitoring.
- Data drift investigation.
- Inference latency measurement.
- Verified maintenance inspection records.

Retraining should be considered when verified performance degradation is observed.

17. Assumptions and Limitations

Assumptions

1. Dataset leakage labels are treated as ground truth.
2. Pressure and refill measurements provide useful information for leakage detection.
3. Previous pressure readings are available to calculate pressure change.
4. The dataset is suitable for an initial experimental study.

Limitations

1. Only three input features were used.
2. Training was performed on a sampled subset.
3. The dataset is simulation-based.
4. The physical sensor system has not been deployed.
5. Hyperparameter tuning was not exhaustive.
6. Inter-tank relationships were not explicitly used as model features.
7. Real-world performance remains unverified.
8. Prediction thresholds have not been optimized using operational cost information.
9. Permutation importance does not establish causal relationships.

18. Future Scope

Future improvements may include:

- Adding inter-tank connectivity features.
- Investigating additional time-series features.
- Applying systematic hyperparameter optimization.
- Evaluating calibrated prediction probabilities.
- Testing with actual water distribution measurements.
- Developing a real-time monitoring interface.
- Integrating physical water tank sensors.
- Implementing automated drift detection.

19. Conclusion

This project investigated supervised machine learning methods for detecting water leakage in a simulated network of connected water tanks.

Four classification models were implemented and evaluated using a chronological sampling strategy.

Random Forest achieved the highest accuracy, while SVM achieved the highest recall and F1-score.

Considering that missed leakage events are particularly important in this application, SVM was selected as the preferred experimental model.

The project also included feature importance analysis, error analysis, simulated inference, deployment documentation and monitoring procedures.

The results demonstrate the potential of machine learning for sensor-based water leakage detection.

However, further validation with real-world sensor measurements is required before operational deployment.

20. References

1. Khorasgani, H., Hasanzadeh, A., Farahat, A. K., & Gupta, C. (2019). Fault Detection and Isolation in Industrial Networks using Graph Convolutional Neural Networks. 2019 IEEE International Conference on Prognostics and Health Management, 1–7. https://doi.org/10.1109/ICPHM.2019.8819403

2. Alves Coelho, J., Glória, A., & Sebastião, P. (2020). Precise Water Leak Detection Using Machine Learning and Real-Time Sensor Data. IoT, 1(2), 474–493. https://doi.org/10.3390/iot1020026

3. Mashhadi, N., Shahrour, I., Attoue, N., El Khattabi, J., & Aljer, A. (2021). Use of Machine Learning for Leak Detection and Localization in Water Distribution Systems. Smart Cities, 4(4), 1293–1315. https://doi.org/10.3390/smartcities4040069

4. Kammoun, M., Kammoun, A., & Abid, M. (2022). Leak Detection Methods in Water Distribution Networks: A Comparative Survey on Artificial Intelligence Applications. Journal of Pipeline Systems Engineering and Practice, 13(3). https://doi.org/10.1061/(ASCE)PS.1949-1204.0000646

5. Shen, Y., & Cheng, W. (2022). A Tree-Based Machine Learning Method for Pipeline Leakage Detection. Water, 14(18), 2833. https://doi.org/10.3390/w14182833

6. Lee, S., & Kim, B. (2023). Machine Learning Model for Leak Detection Using Water Pipeline Vibration Sensor. Sensors, 23(21), 8935. https://doi.org/10.3390/s23218935

21. Reproducibility Information

Dataset repository commit: 862da8b419e65b240c107a343b72dad44915c811

Python: 3.13.18

Pandas: 2.2.3

NumPy: 2.4.6

Scikit-learn: 1.6.1

Random seed used for sampling: 42

Dataset checksums are maintained in: documentation/dataset_manifest.csv

Model configurations are maintained in: documentation/model_configurations.json